NAIVE BAYES CLASSIFIER

“Learning from Probability to Make Smart Predictions”

A Probabilistic Machine Learning Algorithm

Naive Bayes is a supervised machine learning algorithm based on Bayes’ Theorem that predicts a class using probabilities. It assumes that features are conditionally independent given the class.

1 HOW DOES IT WORK?

Input Data
features X

→

Calculate Probabilities
priors & likelihoods

→

Apply Bayes’ Theorem
posterior for each class

→

Compare Class Probabilities
choose the largest

→

Predicted Class

2 THE MATHEMATICAL FORMULA

$$P(C|X) = \frac{P(X|C) \times P(C)}{P(X)}$$

Bayes’ Theorem: class C, observed features X

The “naive” assumption: Features are conditionally independent given the class.

- **P(C|X)** — **Posterior** — probability of class C given the observed features X.
- **P(X|C)** — **Likelihood** — probability of observing features X given class C.
- **P(C)** — **Prior** — probability of class C before observing the features.
- **P(X)** — **Evidence** — probability of observing the features.

For multiple features, the likelihood is commonly modelled as:

$$P(X|C) = P(x_1|C) \times P(x_2|C) \times \ldots \times P(x_n|C)$$

This product follows directly from the conditional-independence assumption.

3 REAL-LIFE EXAMPLE — EMAIL SPAM DETECTION

✉ Incoming Email

“Click now! **Free** **Prize** for the **Winner**!”

SPAM **NOT SPAM**

1. **Examine the features:** the words “Free”, “Prize” and “Winner”.
2. **Estimate likelihoods:** how likely each word is in spam and in non-spam emails (learned from training data).
3. **Calculate class scores:**

Spam: P(Spam) × P(Free|Spam) × P(Prize|Spam) × P(Winner|Spam)

Not Spam: P(Not Spam) × P(Free|Not Spam) × P(Prize|Not Spam) × P(Winner|Not Spam)

4. **Predict the most probable class:** P(X) is the same for both classes, so the class with the larger score wins.

SPAM **Example prediction.** *This is an illustrative example; the prediction depends on probabilities learned from training data.*

4 TYPES OF NAIVE BAYES

Gaussian
Suitable for continuous numerical features commonly modelled using Gaussian distributions.

Multinomial
Commonly used for word counts and text classification.

Bernoulli
Suitable for binary features, such as whether a word is present or absent.

5 ADVANTAGES & LIMITATIONS

Advantages

- ✔ Simple and computationally efficient.
- ✔ Fast to train and predict.
- ✔ Effective for many text-classification tasks.
- ✔ Works well with high-dimensional feature spaces.

Limitations

- ▲ The conditional-independence assumption may be unrealistic.
- ▲ Correlated features can reduce its effectiveness.
- ▲ Zero-frequency feature values can cause zero probabilities; smoothing methods such as Laplace smoothing help address this.
- ▲ Performance depends on the dataset and problem.

6 APPLICATIONS

Email Spam Filtering · **Sentiment Analysis** · **Document Classification** · **Basic Medical Decision Support**

KEY TAKEAWAY: Naive Bayes uses probability and a simplifying independence assumption to classify data efficiently.

Machine Learning | Supervised Learning | Classification